PORTFOLIO PROJECT · 2025

Customer & Marketing Analytics Report

BG-Ride · Data Analytics Division

ROI Attribution

K-Means Segmentation

BG/NBD Churn Model

LTV Analysis

At-Risk Alert

Tools: Python · Pandas · Scikit-learn · PyMC-Marketing (BG/NBD) · Matplotlib · Seaborn · Parquet

Project Overview & Analysis Pipeline

Data: marketing_spend.parquet

1,095 rows

Daily ad spend × 3 channels × 365 days

Data: customer_transactions.parquet

5,000 rows

Orders with amount & channel origin

Analysis Pipeline

1. **Data Load & Quality Check**

   Read Parquet files -> check nulls -> validate column types

2. **ROI Attribution**

   Merge spend + revenue by channel -> Net Profit / Total Spend ratio

3. **RFM + K-Means (K=4)**

   rfm_summary -> Elbow + Silhouette -> 4 clusters -> business labels

4. **Churn Rule-Based**

   last_order_days vs segment threshold (mean + std) -> churned True/False

5. **BG/NBD Model (MAP)**

   BetaGeoModel fit -> probability_alive per customer -> join to segments

6. **At-Risk Alert Export**

   Filter p_alive < 0.3 AND churned == False -> 88 customers -> CSV for CRM

Executive Summary · Key Metrics

2.52M

Total Revenue (2025)

All channels combined

1.42M

Net Profit (2025)

56.2% overall margin

88

At-Risk Customers

P(alive)<0.3, not churned yet

43%

phi_dropout

New customers churn after 1st buy

1.77x

Facebook ROI

Best performing channel

2,039

Platinum LTV

Highest-value segment avg

0.97

Gold P(alive)

Most engaged segment

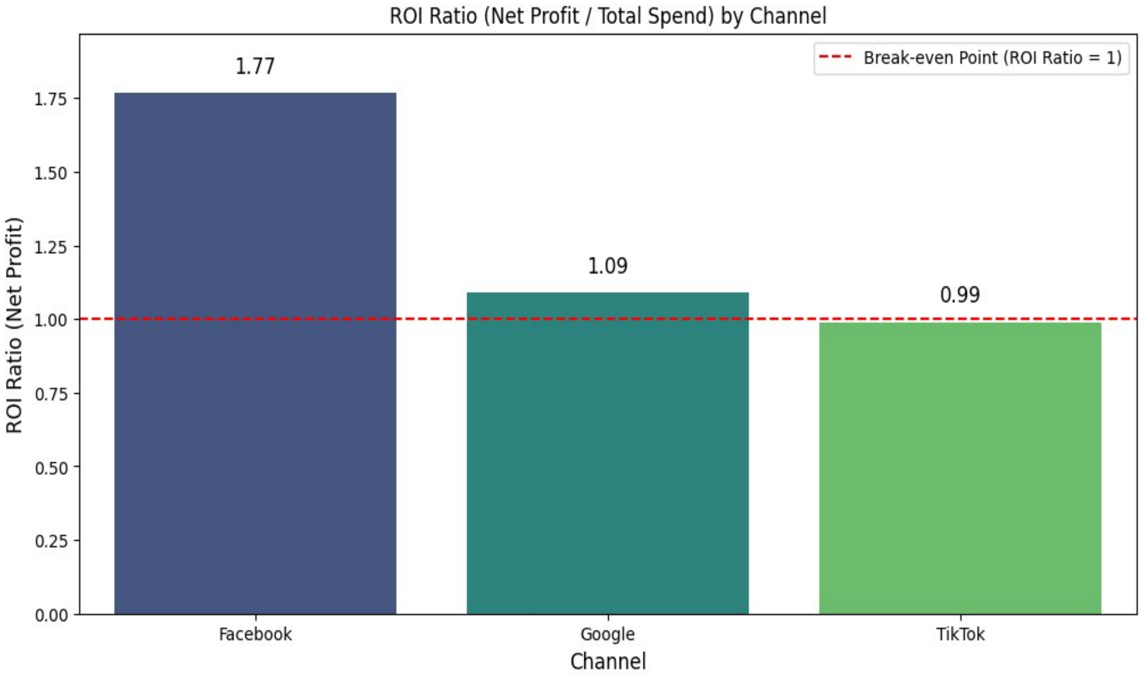
K=4

Optimal Clusters

Elbow + Silhouette validated

Marketing ROI by Channel

ROI = Net Profit / Total Spend



Recommendation: Shift TikTok budget to Facebook immediately. TikTok ROI at 0.99x means every ad Baht spent generates negative return. Optimize Google bidding before scaling.

Channel Breakdown

Channel	ROI	Verdict
Facebook	1.77x	Best
Google	1.09x	Viable
TikTok	0.99x	At Loss

Full-Year 2025

Total Revenue:	2,523,340
Total Spend:	1,105,185
Net Profit:	1,418,155
Margin:	56.2%

Customer Segmentation · K-Means (K=4)

Features Used (StandardScaler)

- AVG_amount — Average order value per customer
- transaction_count — Purchase frequency
- LTV — (freq+1) x monetary x T / 360

Why K=4? (from notebook)

Elbow Method -> inflection point clearly at K=4

Silhouette Score -> K=4 scored best (closest to 1.0)

Business Logic -> 4 tiers align with marketing budget tiers

Platinum

High Value

Avg LTV: 2,039

Profile: High spend, high freq

Action:

VIP retention, loyalty rewards, personal offers

Gold

Mid-High Value

Avg LTV: 31

Profile: Mid-high, good freq

Action:

Upsell campaigns -> push to Platinum

Silver

Mid Value

Avg LTV: 476

Profile: Mid spend, lower freq

Action:

Re-engagement offers, increase purchase freq

Bronze

Low Value

Avg LTV: 0

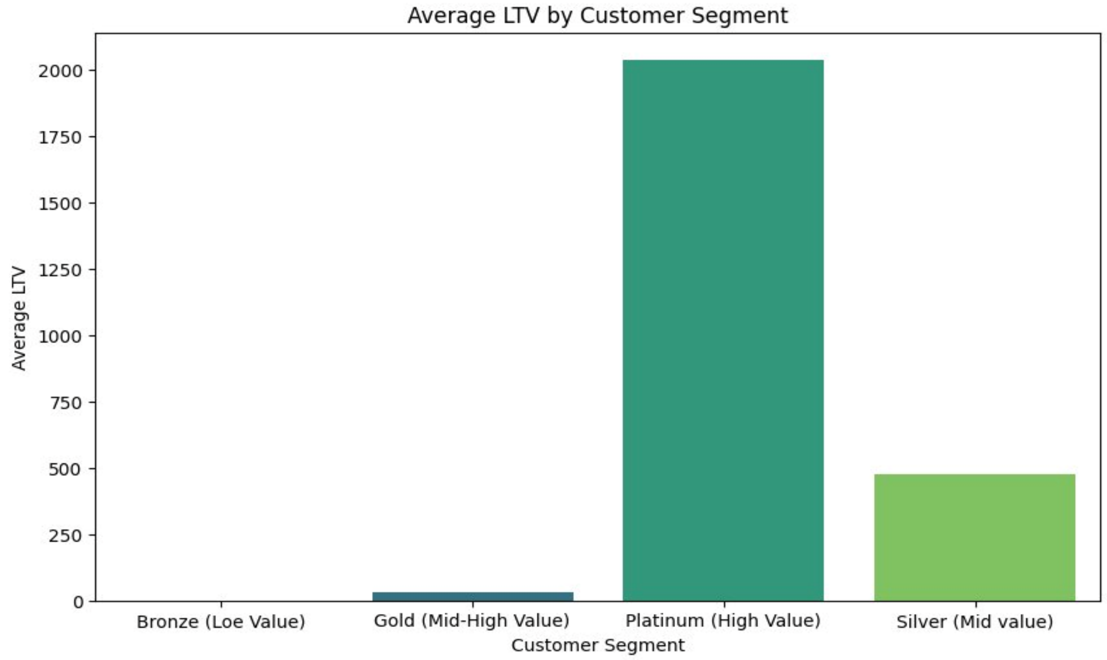
Profile: Low spend, low freq

Action:

Volume play, activation, referral programs

Lifetime Value (LTV) by Segment

(freq+1) x monetary x T / 360



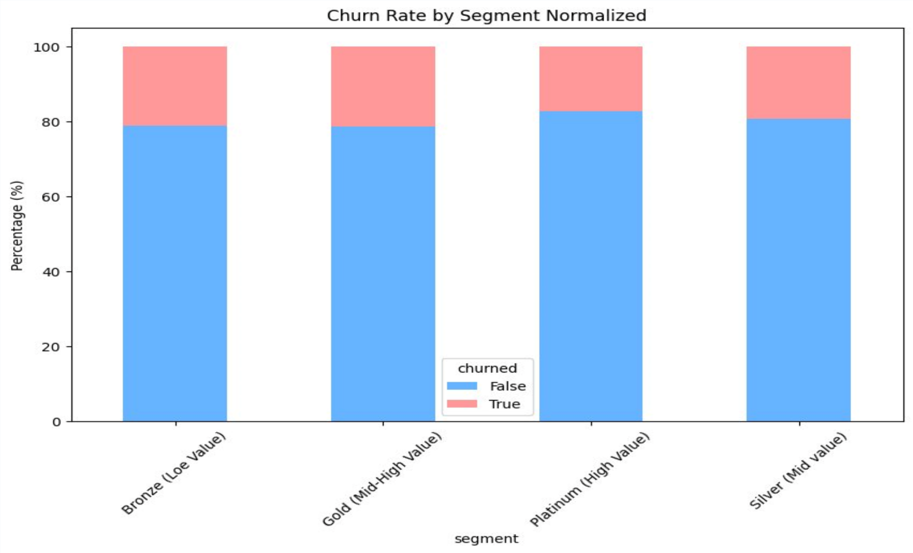
* LTV formula from notebook Cell 18: (frequency+1) x monetary_value x T / 360

LTV Stats by Segment

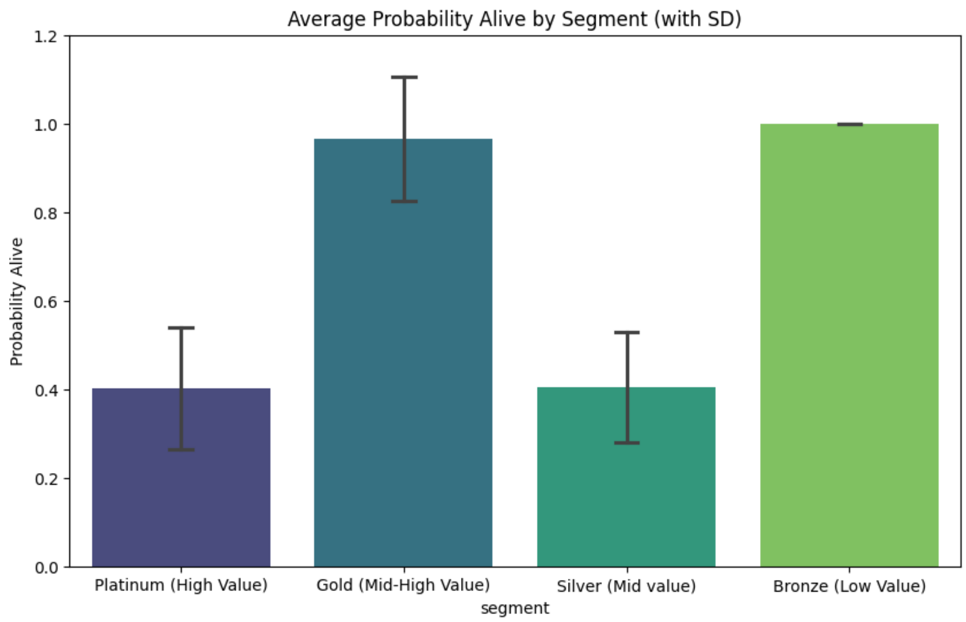
Segment	Mean	Median
Platinum	**2,039**	1,785
Silver	**476**	400
Gold	**31**	0
Bronze	**0**	0

Platinum LTV is 66x higher than Gold. Retaining 1 Platinum customer = retaining 66 Gold customers in revenue terms.

Churn Analysis · Rule-Based + BG/NBD Model



Method 1: Rule-Based (last_order_days > mean+std per segment)



BG/NBD Model Parameters (MAP Estimation) · Model fit via PyMC-Marketing BetaGeoModel

- r=0.457, alpha=89.86 — Purchase behavior: high variation in rates; long average intervals between orders
- a=9.13, b=12.11 — Dropout: b>a means customers mostly stay loyal — churn is not immediate
- phi_dropout=0.43 — 43% of new buyers are 'one-hit wonders' — they churn right after 1st purchase
- MAP vs MCMC — Used MAP for speed (point estimate). MCMC would add confidence intervals but takes much longer

At-Risk Customer Alert · 88 Customers Need Action

Filter (notebook Cell 68): `probability_alive < 0.30 AND churned == False`

Active customers showing churn signals — still reachable. This is the intervention window.

Platinum

CRITICAL

High Value

25

customers at risk

Recommended Action:

Immediate personal outreach via phone or account manager. Offer exclusive VIP renewal deal. Each customer = 2,039 LTV at risk.

Silver

HIGH

Mid Value

60

customers at risk

Recommended Action:

Personalized email sequence with win-back coupon. Set 24-48hr expiry to create urgency. Largest group — automate the outreach flow.

Gold

MODERATE

Mid-High Value

3

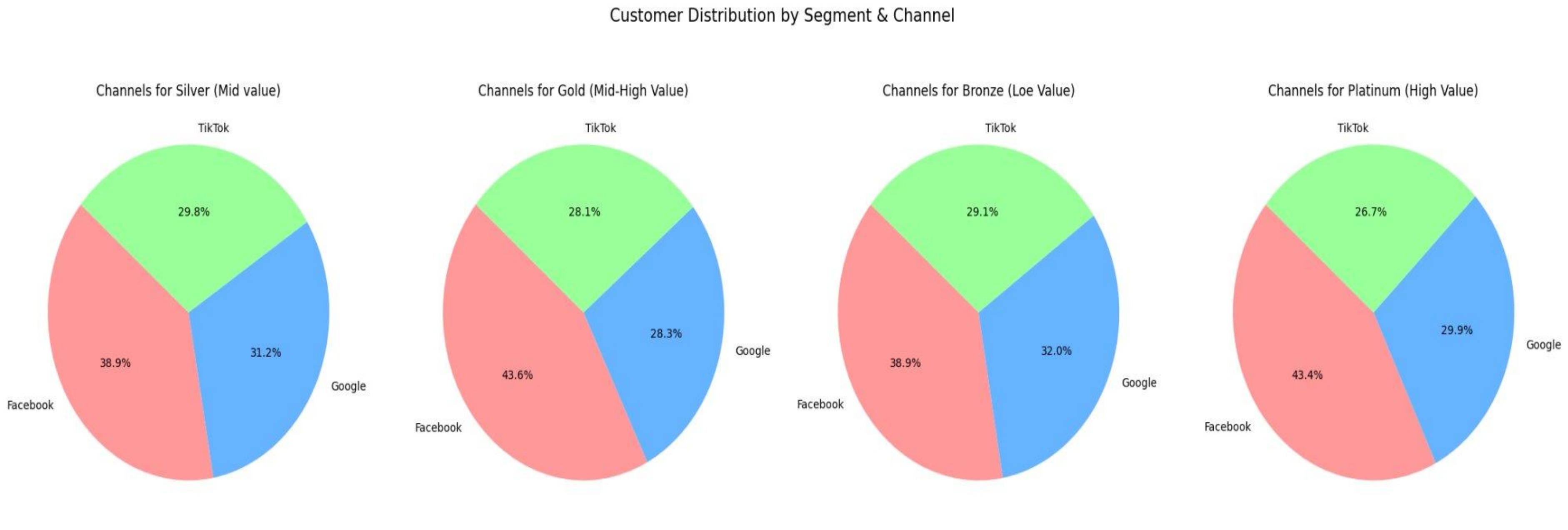
customers at risk

Recommended Action:

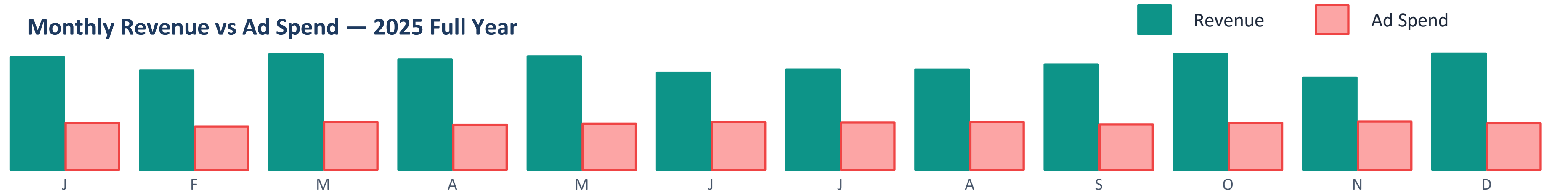
Re-engagement push notification. Highlight new features or limited promotions. Small group — quick manual check sufficient.

Total: 88 at-risk customers | Exported: filtered_customers_alert.csv | Mean P(alive): 0.25 | Avg last_order_days: 162

Customer Distribution by Segment & Channel



Monthly Revenue vs Ad Spend — 2025 Full Year



Conclusions & Next Steps

01 Scale Facebook, Pause TikTok

Facebook ROI=1.77x is clearly the best channel. Reallocate TikTok budget immediately. Only return to TikTok after creative optimization proves ROI>1.0.

02 Emergency: Retain Platinum

25 Platinum customers at critical churn risk. LTV 2,039 each. Deploy VIP retention NOW: personal call, exclusive deal, dedicated manager.

03 Fix phi_dropout=43%

43% of new customers churn after first purchase (BG/NBD finding). Build structured onboarding: follow-up message + offer within 7 days of signup.

04 Silver Group: 60 At-Risk

60 Silver customers need win-back campaign. Automate email sequence with limited-time coupon. This is the highest-volume intervention opportunity.

05 Automate Alert System

filtered_customers_alert.csv is ready for CRM. Next: schedule weekly auto-filter + trigger outreach automatically for scale.

06 Add Cohort Retention Analysis

Notebook identifies this as a gap (Cell 72). Cohort analysis complements BG/NBD — shows time-based retention curves for each acquisition month.